PepXPro

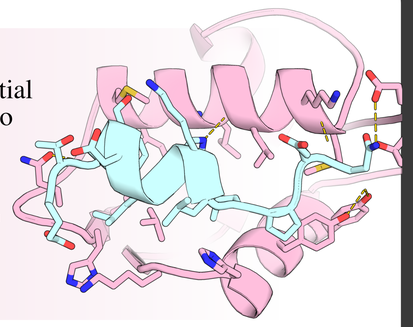
A Framework to Build Reliable Protein-Peptide Datasets for Computational Modeling

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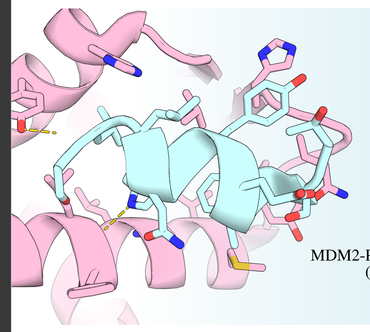
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1 Motivation

Biological relevance
Protein-peptide interactions are essential to cellular processes, from signaling to gene regulation.



MDM2-p53 peptide[1]
(PDB: 1YCR)



MDM2-PMI designed peptide[1]
(PDB: 3EQS)

Therapeutic potential
Peptides can target large, flat, and shallow interfaces that are often considered "undruggable".

Growing Computational Demand. Computational modeling of protein-peptide interactions is rapidly advancing, particularly through **ML-based approaches** for applications such as binding affinity prediction[2,3]. High-quality experimental **structural** and binding **affinity** data are essential for **method development and benchmarking**.

Challenges in Existing Protein-Peptide Resources

Databases
Prioritize comprehensive coverage but require task-specific curation.

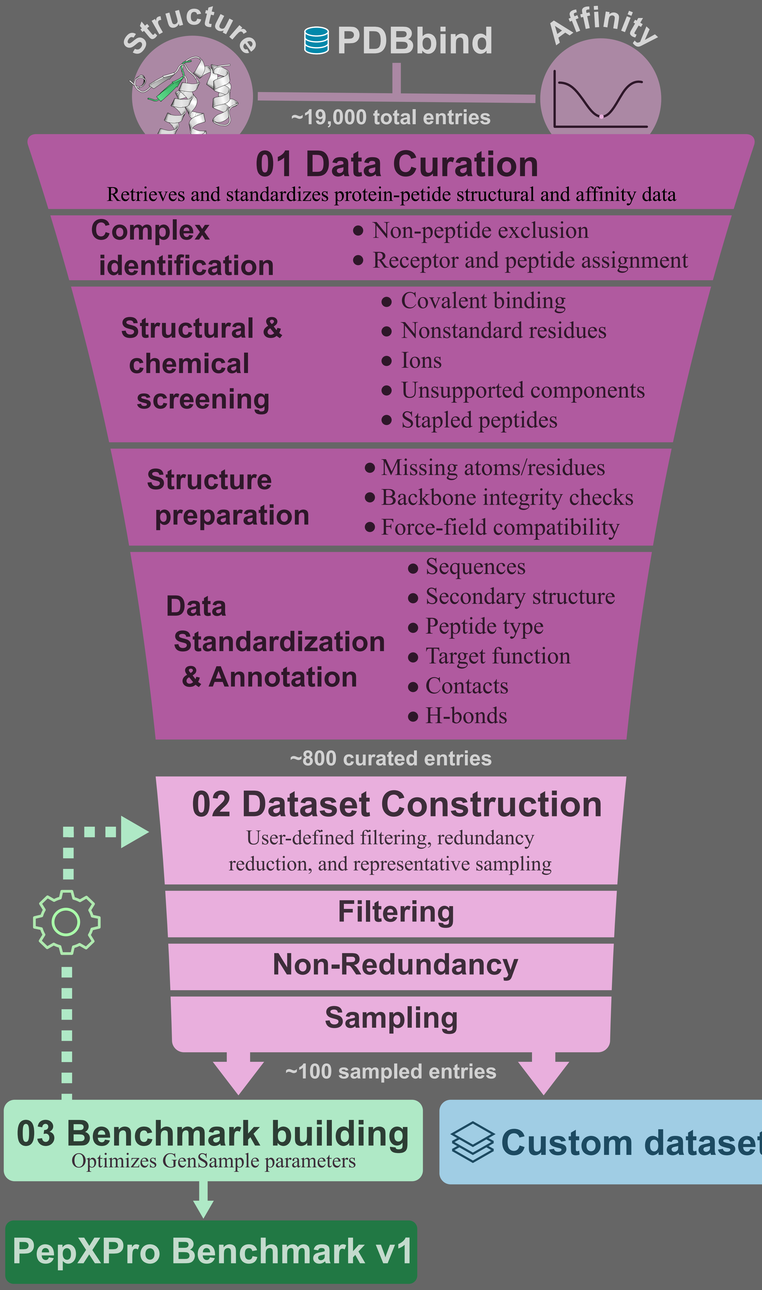
PDBbind
Structural artifacts·High redundancy·Biased affinity distributions[4].

Published benchmarks
Static collections. Heterogeneous curation: quality-filtering·Redundancy-reduction·Sampling strategies[5].

Goal: Develop a reproducible and open-source framework that **bridges** comprehensive protein-peptide **databases** and task-specific **benchmarks** through **curation and customizable dataset generation**.

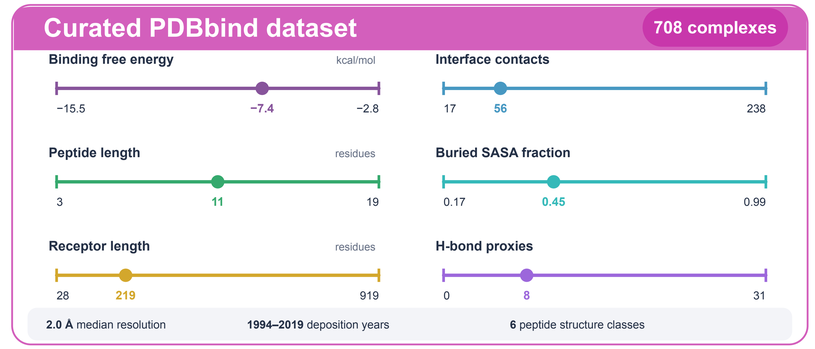
2 PepXPro workflow

PepXPro comprises three modules: **01) Scrape**, for automated protein–peptide data curation; **02) GenSample**, for generating customizable datasets with quality, redundancy, and sampling controls; and **03) Benchmark**, for evaluating and selecting representative, nonredundant benchmark datasets.

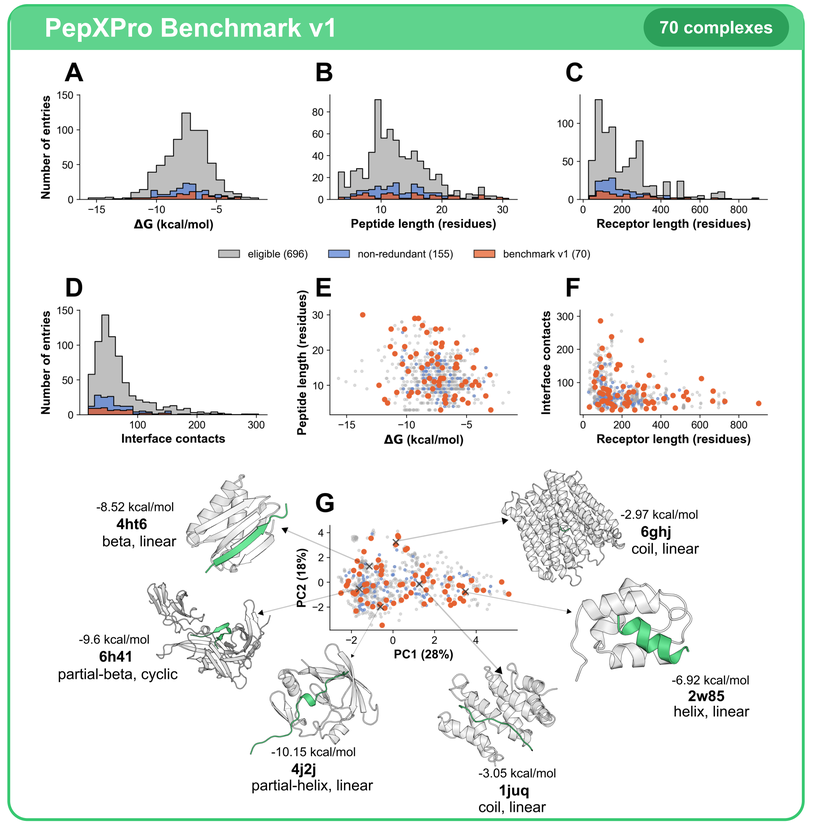


3 Results

The curated pool spans a broad range of binding affinities, molecular sizes, and interface properties.



We generated PepXPro Benchmark v1, a representative, nonredundant dataset of 70 protein-peptide complexes with binding affinities.



Summary & Future work.

- We developed PepXPro, an open-source framework for automated, reproducible curation and customizable dataset generation, retaining diverse systems for structural biology and drug discovery applications.
- Future work includes expanding and continuously updating PepXPro datasets as new high-quality experimental structures and binding affinity data become available.

References
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[3] Moody, M. T., et al. (2026). Biophysical Journal, 125(4), 426a.
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Resources
PepXPro
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